

Chapter ~ 5

Magnetism & Matter.

⇒ Magnetic dipole moment of a current loop.

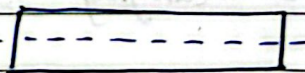
⇒ Magnetic dipole moment of a revolving electron.

Note: These both topics are same as Chapter 4 : Torque and ~~electron~~ as magnetic dipole.

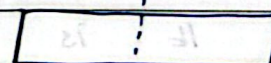
⇒ Bar Magnet

All bar magnets have two poles, the north pole and the south pole. It will always have it.

⇒ Cutting a Bar Magnet.

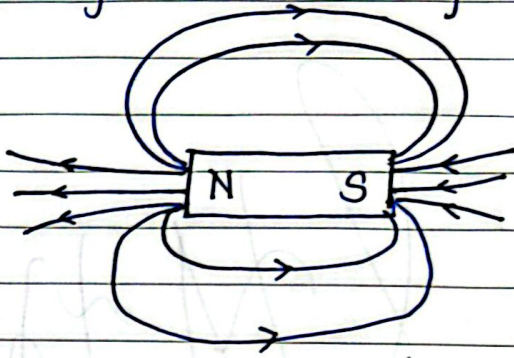
i) Along its length : 

Pole strength reduces by halves
 Length remains same.

ii) Perpendicular its length : 

Pole strength remains same
 Length reduces by half.

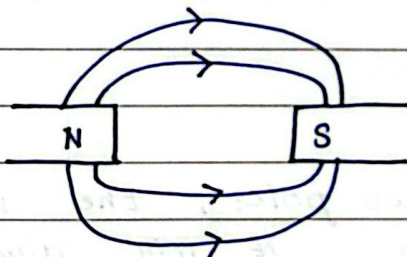
⇒ Magnetic Lines of Force.



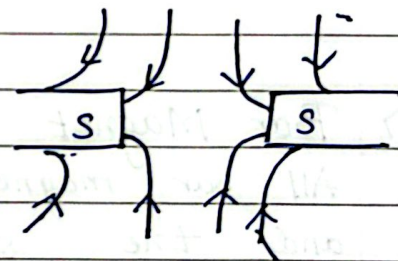
the lines of force in a magnetic field are those imaginary lines which continuously represent the direction of magnetic field.

Properties .

- i) The magnetic lines emerge from north pole and following curved path reach south pole.
- ii) Two lines of force never intersect each other.
- iii) Near the magnetic poles where field is stronger the lines of force are closer.
- iv) No lines of force will pass through neutral point.



(a)



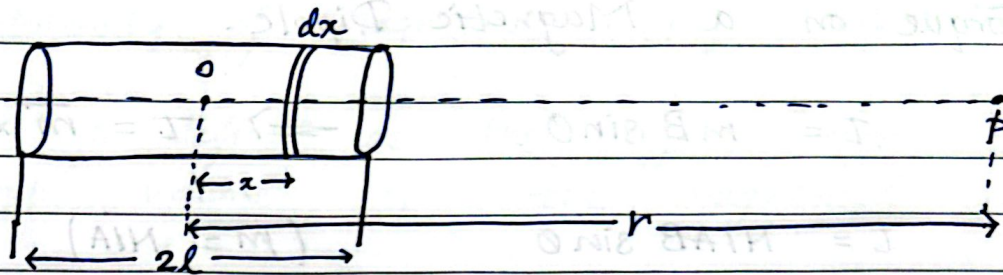
(b)

⇒ A current carrying Solenoid : A Bar Magnet.

→ It always rests in the north-south direction.

→ It is just like a bar magnet having north and south poles.

→ Two if placed will have attraction or repulsion.



⇒ Magnetic Field of a magnetic dipole / Small bar magnet.

i) End on Position:

$$dB = \frac{\mu_0}{4\pi} \frac{2\pi (ndx) Ia^2}{[a^2 + (r-x)^2]^{3/2}}$$

$$\int dB = \frac{\mu_0}{4\pi} 2\pi n Ia^2 \int_{-l}^l \frac{dx}{[a^2 + (r-x)^2]^{3/2}}$$

$$= \frac{\mu_0}{4\pi} 2\pi n Ia^2 \int_{-l}^l \frac{-x' dx}{r^3}$$

$$= \frac{\mu_0}{4\pi} \frac{2\pi n Ia^2}{r^3} (l+l)$$

$$= \frac{\mu_0}{4\pi} \frac{2\pi n Ia^2}{r^3} \times 2l$$

$$m = nIA^2 \times 2l$$

$$B = \frac{\mu_0}{4\pi} \frac{2m}{r^3}$$

ii) Broad side on Position: ($l \ll r$)

$$B = \frac{\mu_0}{4\pi} \frac{m}{r^3}$$

Note: $B_{end} = 2 \times B_{broad}$.

⇒ Torque on a Magnetic Dipole.

$$\tau = mB \sin \theta$$

$$\Rightarrow \tau = \vec{m} \times \vec{B}$$

$$\tau = N I A B \sin \theta$$

$$(m = N I A)$$

	Diamagnetic Substances	Paramagnetic Substances	Ferromagnetic Substances
⇒	feebly magnetised opp to direction of magnetic field	feebly magnetised in the direction of magnetic field	strongly magnetised in the direction of magnetic field.
⇒	they are repelled away from a magnet.	they are attracted towards a magnet.	they are attracted fast towards a magnet.
⇒	Examples: Bismuth, zinc, copper, silver, lead, water, mercury, NaCl, nitrogen, hydrogen	Aluminium, sodium, platinum, manganese, antimony, CuCl_2 , liquid oxygen	Iron, Nickel, Cobalt, magnetite (Fe_3O_4) etc.

As a matter of fact: every substance is in general diamagnetic.

⇒ Some Important Terms used in Magnetism

ii) Magnetic Induction (B): When any substance is placed in an external magnetic field the magnetism produced is induced magnetism.

The number of magnetic lines of induction inside a magnetised substance crossing unit area is called magnitude of magnetic Induction.

ii] Intensity of Magnetisation (M)

It is defined as the magnetic moment per unit volume of the magnetised substance.

$$M = \frac{m}{V} \rightarrow \begin{array}{l} \text{magnetic moment} \\ \text{Volume} \end{array}$$

iii] Magnetic Intesity / Magnetic Field Strength (H).

$$H = \frac{B}{\mu_0} - M.$$

It is the capability of magnetising field to magnetise the substance.

iv] Magnetic Permeability (μ)

It is defined as the ratio of magnetic Induction B inside the magnetised substance to the magnetic intensity (H)

$$\mu = \frac{B}{H}.$$

v] Relative Magnetic Permeability (μ_r)

$$\mu_r = \frac{\mu}{\mu_0} \rightarrow \begin{array}{l} \text{magnetic permeability} \\ \text{magnetic permeability in free space.} \end{array}$$

$$\mu_r = \frac{B}{B_0}.$$

vi) Magnetic Susceptibility (χ_m)

It is defined as the ratio of the intensity of magnetisation to the magnetic intensity of magnetising field.

$$\chi_m = \frac{M}{H}$$

χ_m of Diamagnetic $\Rightarrow$ small and negative ($\chi_m < 0$)

χ_m of Paramagnetic $\Rightarrow$ small positive values ($\chi_m > 0$)

χ_m of Ferromagnetic $\Rightarrow$ very large +ve ($\chi_m \gg 0$).

$\Rightarrow$ Relation between relative permeability (μ_r) and Magnetic Susceptibility (χ_m).

$$H = \frac{B}{\mu_0} - M$$

$$H + M = \frac{B}{\mu_0}$$

$$\mu_0 (H + M) = B$$

$$\mu_0 (H + M) = \mu H$$

$$\mu_0 \left(1 + \frac{M}{H}\right) = \mu$$

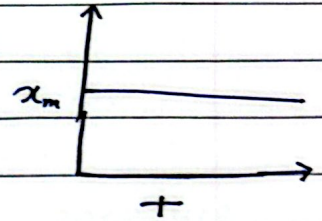
$$i) \quad \boxed{\mu_0 (1 + \chi_m) = \mu}$$

$$(1 + \chi_m) = \frac{\mu}{\mu_0}$$

$$ii) \quad \boxed{1 + \chi_m = \mu_r}$$

⇒ Effect of Temperature on Magnetic Materials.

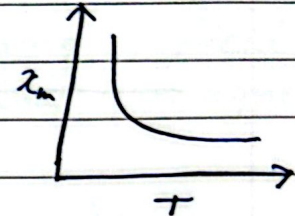
- 1) Diamagnetic Materials : It is independent of kelvin Temperature.



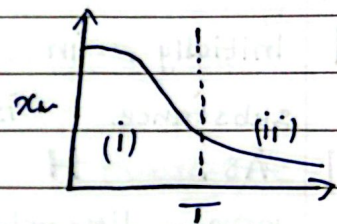
- 2) Paramagnetic Materials : It is inversely proportional to temp.

$$\chi_m = \frac{C}{T}$$

Curie Constant.



- 3) Ferromagnetic Materials : Temperature at which ferromagnetic material behaves as a paramagnetic material is called Curie's Temperature.

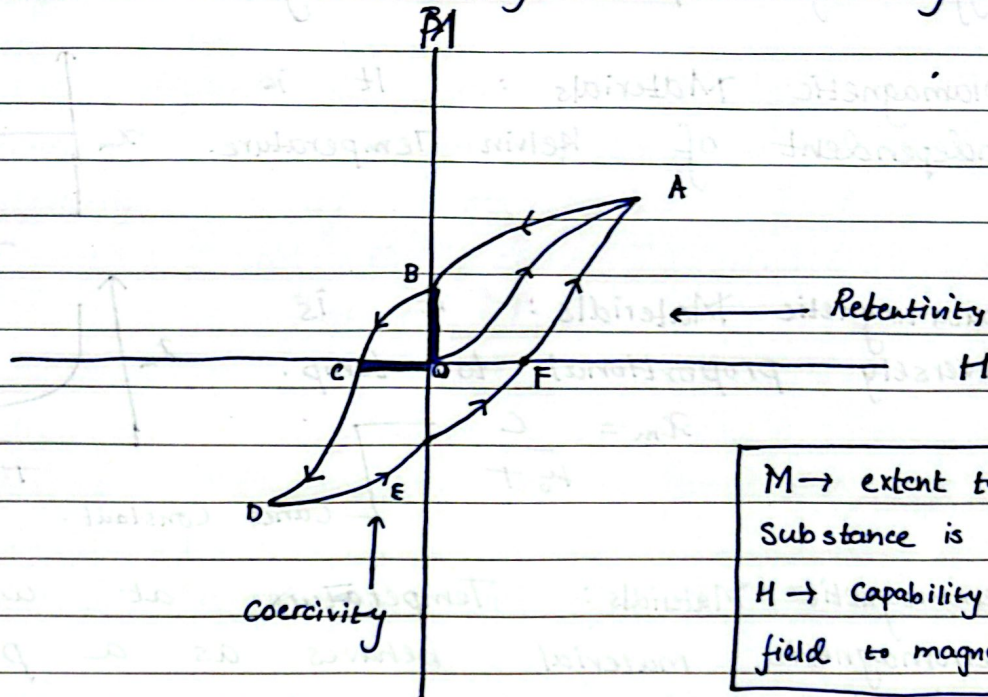


⇒ Curies Laws

It states that $\chi_m \propto 1/T$
 Magnetic susceptibility of a paramagnetic material is inversely proportional to the Temperature.

$$\chi_m \propto \frac{1}{T}$$

⇒ Hysteresis : Retentivity and Coercivity.



- i] Initially, in Absence of Magnetic Field the substance is in unmagnetised state.
- ii] As, H is increased, M also increases non-linearly (point A is state of magnetic saturation of the substance).
- iii] Any further increase in H does not produce any increase in M .
- iv] If now the magnetising field H is decreased M also decreases along path AB, thus M lags behind. when H becomes 0 (at point B) M has still value OB.
- v] The magnetisation remaining in the substance (when M is reduced to 0) is called residual Magnetism.

Resistivity - It is the measure of magnetisation remained in the substance when $H=0$

vi] If now H is increased in the reverse direction, M decreases along BC until it becomes 0 at C
 (At point C $[M=0, H \neq 0]$)

Point C $\Rightarrow$ Coercivity.

vii] $\Rightarrow$ Coercivity: It is the measure of the reverse magnetising field required to destroy residual magnetism of substance.

vii] As H is increased beyond OC the substance is magnetised in the opp. direction along CD. at D substance is again saturately magnetised.

viii] By taking H from its negative value to its original maxim value of symmetrical curve DEFA.

ix] At points B and E substance is magnetised in absence of magnetic field.

The cycle of ABCDEFA is called cycle of magnetisation.

$$\chi_m = \frac{H}{M}$$

Hysteresis loss: Area under loop gives hysteresis loss.

The energy lost per unit volume of a substance in a complete cycle of magnetisation which is equal to the area of hysteresis loop is called hysteresis loss.

⇒ Difference in magnetic properties of soft iron and steel.

retentivity soft iron $>$ retentivity of steel

coercivity soft iron $<$ coercivity of steel.

∴ Hysteresis loss in soft iron is smaller than that in steel.

Note: Material with minimum loss is most suitable.

⇒ Selection of magnetic materials

i] Permanent magnets : high retentivity
 high coercivity

ii] Electromagnets : Low retentivity
 (soft iron).

iii] Transformers : Low Hysteresis Loss.